



Retail Sales Data Cleaning & Analysis

In this project, I worked with a raw transactional sales dataset containing product names, prices, quantities, payment methods, and locations. The goal was to clean the data and generate business-relevant insights using visualizations.

Key Objectives:

- Clean and prepare the data for analysis
- Understand monthly sales trends
- Analyze product performance
- Explore customer payment behavior
- Discover location-wise patterns

IMPORTING LIBRARIES

```
In [3]: import numpy as np
import pandas as pd
import warnings
df = pd.read_csv(r"C:\Users\Aamir Bin Raheem\Downloads\practice datasets\dirty_cafe_sales.csv")
warnings.filterwarnings('ignore')
```

```
In [5]: # checking for total number of null values in each column
df.isna().sum()
```

```
Out[5]: Transaction ID      0
        Item              333
        Quantity          138
        Price Per Unit     179
        Total Spent        173
        Payment Method     2579
        Location           3265
        Transaction Date    159
        dtype: int64
```

USING A FUNCTION TO CLEAN AND IMPUTE COLUMN VALUES

```
In [8]: def clean_column(col, strategy='mean'):
        df[col] = df[col].apply(lambda x: np.nan if str(x).strip().upper() in ['UNKNOWN', 'ERROR'] else x)
        if strategy == 'mean':
            df[col] = pd.to_numeric(df[col], errors='coerce')
            df[col].fillna(df[col].mean(), inplace=True)
        elif strategy == 'mode':
            df[col].fillna(df[col].mode()[0], inplace=True)

        # so instead of cleaning the columns one by one, we can simply use this function
```

```
In [10]: df.columns
```

```
Out[10]: Index(['Transaction ID', 'Item', 'Quantity', 'Price Per Unit', 'Total Spent',
               'Payment Method', 'Location', 'Transaction Date'],
              dtype='object')
```

DATA CLEANING

```
In [13]: clean_column('Quantity', 'mean')
        clean_column('Item', 'mode')
        clean_column('Price Per Unit', 'mean')
        clean_column('Payment Method', 'mode')
        clean_column('Location', 'mode')
        clean_column('Transaction Date', 'mode')
```

```
In [15]: # checking for missing values after imputations
df.isna().sum()
```

```
Out[15]: Transaction ID      0
Item                        0
Quantity                   0
Price Per Unit             0
Total Spent                173
Payment Method             0
Location                   0
Transaction Date           0
dtype: int64
```

```
In [17]: # converting ERROR and UNKNOWN values in 'Total Spent' column to NaN
df["Total Spent"] = df["Total Spent"].apply(lambda x: np.nan if str(x).strip().upper() == "UNKNOWN" else x)
df["Total Spent"] = df["Total Spent"].apply(lambda x: np.nan if str(x).strip().upper() == "ERROR" else x)
df["Total Spent"].isna().sum()
```

```
Out[17]: 502
```

```
In [19]: # Handling NaN values in 'Total Spent' column
df["Total Spent"] = df.Quantity * df["Price Per Unit"]
df["Total Spent"].isna().sum()
```

```
Out[19]: 0
```

VISUALIZATION

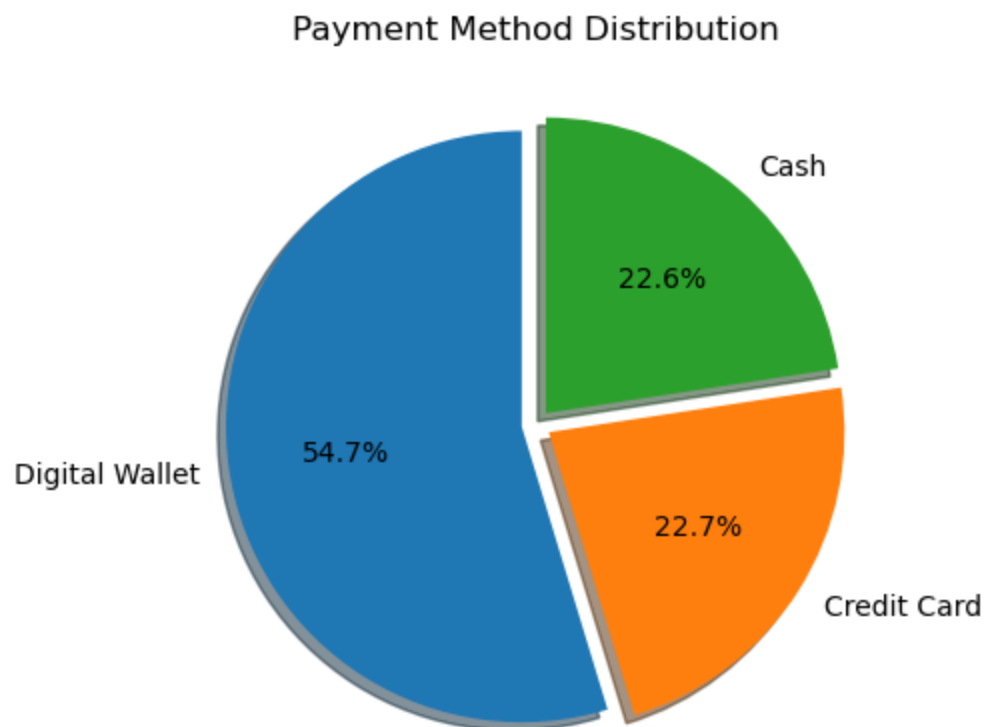
```
In [22]: import matplotlib.pyplot as plt
```

1. WHAT IS THE PAYMENT METHOD DISTRIBUTION?

```
In [25]: df['Payment Method'].value_counts().plot(
    kind = 'pie',
    autopct = '%1.1f%',
    startangle = 90,
    explode = [0.05]*df['Payment Method'].nunique(),
```

```
shadow = True,  
title = 'Payment Method Distribution',)  
plt.ylabel('')
```

Out[25]: Text(0, 0.5, '')



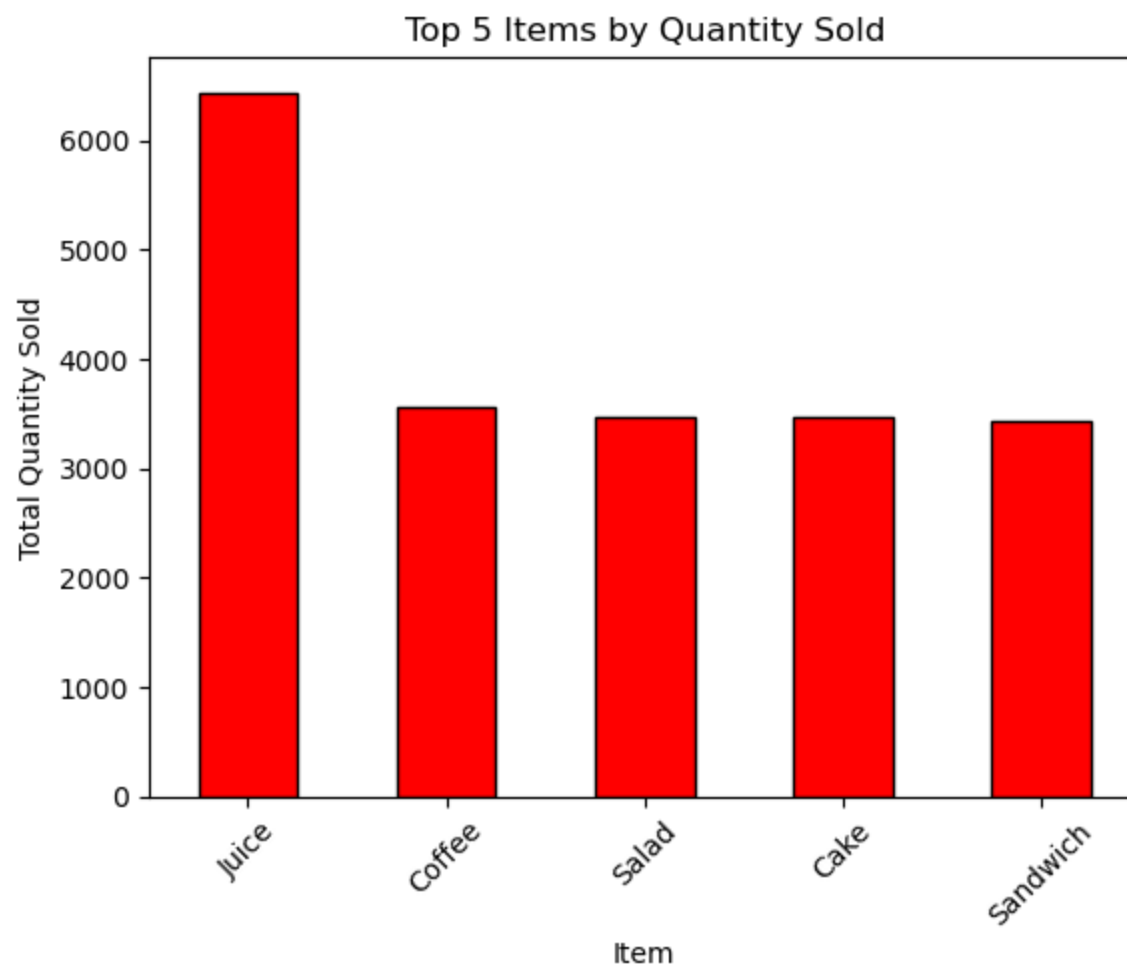
```
In [ ]: '''The payment Method distribution shows that the majority of transactions were made using Digital Wallets,  
followed by Credit Cards and Cash.'''
```

2. WHAT ARE THE TOP 5 ITEMS BY QUANTITY SOLD?

```
In [27]: top_items = df.groupby("Item")["Quantity"].sum().sort_values(ascending=False).head(5)  
  
top_items.plot(  
    kind = 'bar',
```

```
    title = 'Top 5 Items by Quantity Sold',  
    color = 'Red',  
    edgecolor = 'Black',  
    )  
plt.xlabel('Item')  
plt.ylabel('Total Quantity Sold')  
plt.xticks(rotation=45)  
plt.xticks(rotation=45)
```

```
Out[27]: (array([0, 1, 2, 3, 4]),  
 [Text(0, 0, 'Juice'),  
  Text(1, 0, 'Coffee'),  
  Text(2, 0, 'Salad'),  
  Text(3, 0, 'Cake'),  
  Text(4, 0, 'Sandwich')])
```



```
In [ ]: '''Top 5 Items by Quantity sold are Juice, Coffee, Salad, Cake and Sandwich'''
```

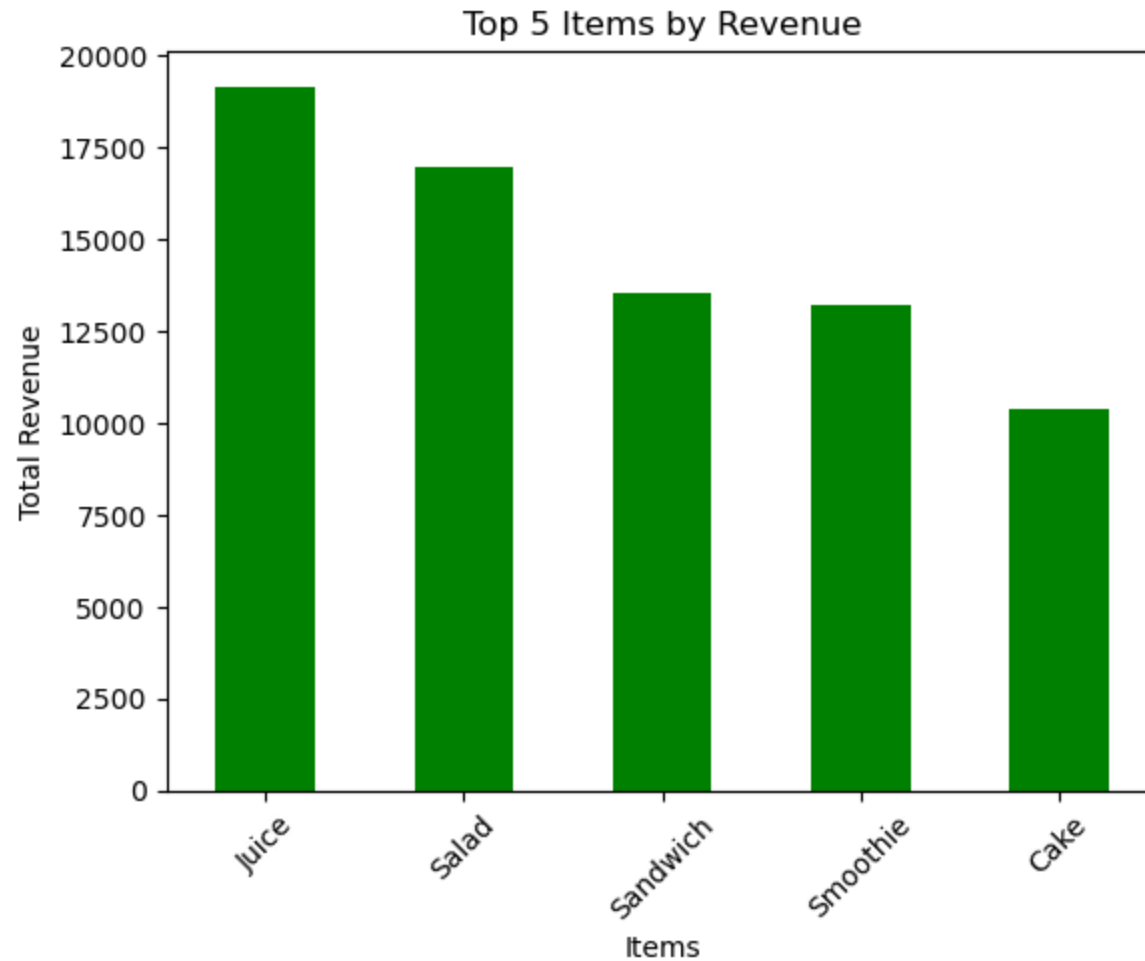
3. WHAT ARE THE TOP 5 ITEMS WITH MOST REVENUE?

```
In [29]: most_revenue = df.groupby("Item")["Total Spent"].sum().sort_values(ascending = False).head().round(2)

most_revenue.plot(
    kind = 'bar',
    title = 'Top 5 Items by Revenue',
```

```
color = 'Green'  
)  
plt.xticks(rotation = 45)  
plt.xlabel("Items")  
plt.ylabel("Total Revenue")
```

Out[29]: Text(0, 0.5, 'Total Revenue')

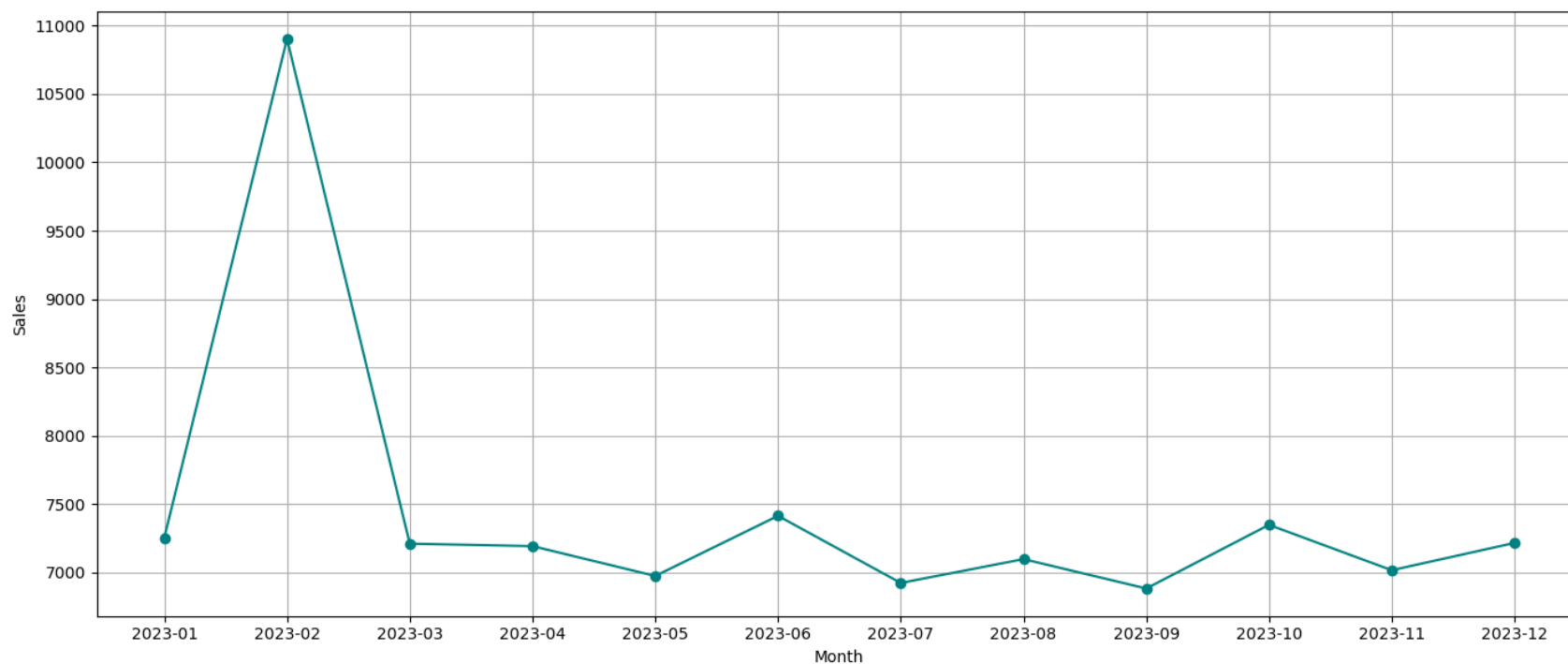


4. WHAT'S THE OVERALL SALES TREND OVER TIME?

```
In [31]: # creating a new column called 'Month'
df["Transaction Date"] = pd.to_datetime(df["Transaction Date"])
df["Month"] = df["Transaction Date"].dt.to_period('M')

In [33]: # Grouping by 'Month' and summing 'Total Spent' to get monthly sales trends
Trends = df.groupby("Month")["Total Spent"].sum().sort_index(ascending = True)

plt.figure(figsize=(14,6))
plt.plot(Trends.index.astype(str), Trends.values, color = 'teal', marker = 'o')
plt.xlabel("Month")
plt.ylabel('Sales')
plt.tight_layout()
plt.grid(True)
plt.show()
```



```
In [ ]: '''As you can see, the sales trend gradually increased in January and peaked in the month of February,
and gradually declined in the month of March. Following that, there have been minimal fluctuations'''
```

5. WHICH LOCATIONS PREFER WHICH PAYMENT METHODS?

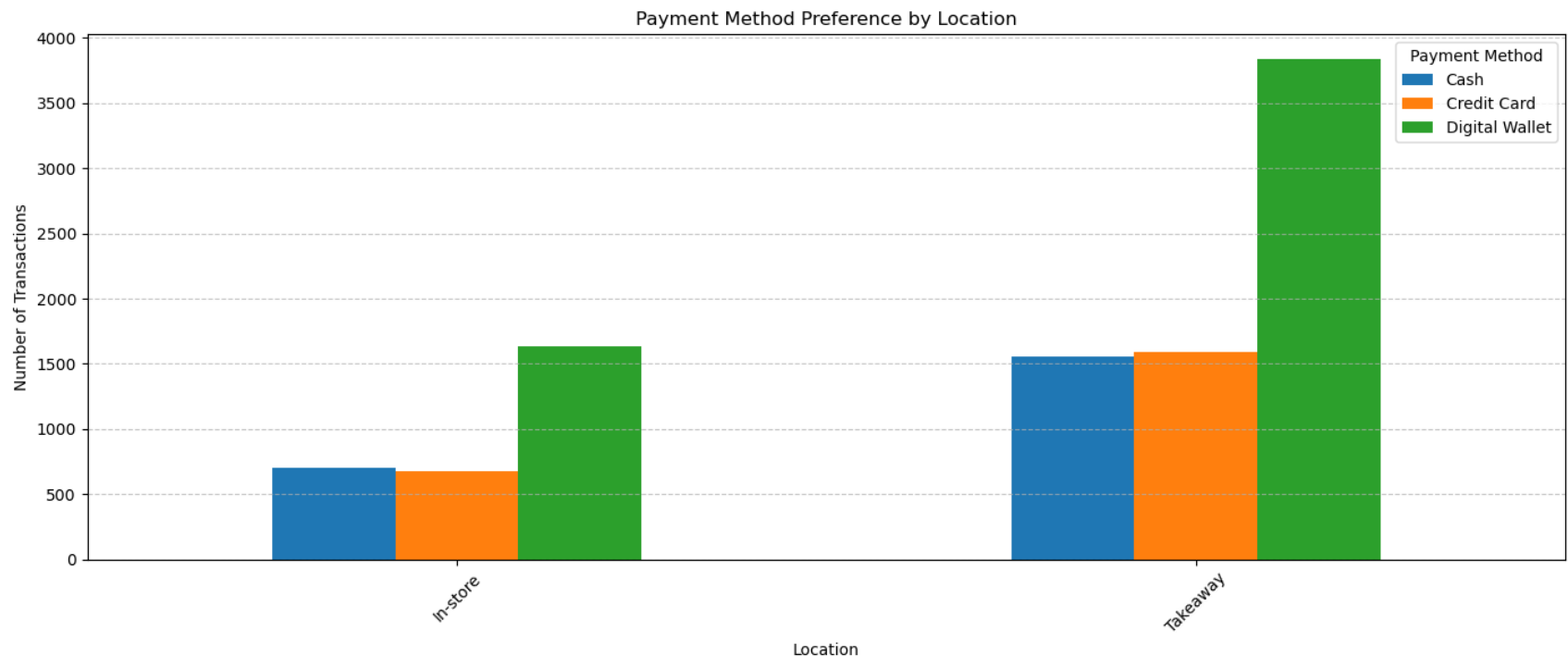
```
In [35]: Preference = df.groupby(['Location', 'Payment Method']).size().unstack(fill_value=0)
Preference
```

```
Out[35]: Payment Method  Cash  Credit Card  Digital Wallet
```

Location			
In-store	702	681	1634
Takeaway	1556	1592	3835

```
In [42]: # Plot
Preference.plot(kind='bar', figsize=(14,6))

plt.title('Payment Method Preference by Location')
plt.xlabel('Location')
plt.ylabel('Number of Transactions')
plt.xticks(rotation=45)
plt.legend(title='Payment Method')
plt.tight_layout()
plt.grid(axis='y', linestyle='--', alpha=0.7)
plt.show()
```



```
In [ ]: '''Based on the bar chart, it is evident that both locations prefer Digital Wallets as their primary payment method'''
```



Conclusion

This project offered a comprehensive look into customer transactions. Some key insights include:

- February had the highest sales — further investigation is needed to understand the spike.
- Digital Wallets are the most preferred payment method, especially for takeaway orders.
- Juice leads in quantity sold, while Coffee dominates in total revenue.

By combining data cleaning and Python-based visualizations, I was able to transform raw data into actionable insights. This project demonstrates how even a small dataset can reveal important business patterns when analyzed the right way.

```
In [ ]:
```